

Drug Therapy Quick Reference

A concise clinical reference of common medications, dosing, and safety monitoring.

NOTE This is a **synthetic document created for a software demonstration**. It is not medical advice and must not be used for patient care. Citations are illustrative.

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1. Warfarin

Oral anticoagulant — vitamin K antagonist

KEY FACT Target INR for most indications is 2.0–3.0; mechanical mitral valves target 2.5–3.5 [1].

Mechanism of action

Warfarin inhibits vitamin K epoxide reductase (VKORC1), reducing synthesis of clotting factors II, VII, IX, and X. Onset is delayed 24–72 hours because circulating factors must first clear [1].

Dosing

Setting	Dose
Initiation (typical)	5 mg once daily, then adjust to INR
Reversal (major bleed)	Vitamin K + 4-factor PCC [2]

Safety and monitoring

Dosing is highly individual: **CYP2C9** and **VKORC1** genotype, diet (vitamin K intake), and many drug interactions shift the INR. Check INR frequently after any regimen change [1].

CLINICAL PEARL Amiodarone, fluconazole, and metronidazole raise the INR and often require a warfarin dose reduction [2].

2. Apixaban

Direct oral anticoagulant (DOAC) — factor Xa inhibitor

KEY FACT Standard dose in atrial fibrillation is 5 mg twice daily; reduce to 2.5 mg twice daily if the patient has at least two of: age $\geq$ 80 years, weight $\leq$ 60 kg, or serum creatinine $\geq$ 1.5 mg/dL [3].

Mechanism of action

Apixaban directly and reversibly inhibits factor Xa, blocking conversion of prothrombin to thrombin. No routine INR monitoring is required [3].

Dosing

Setting	Dose
Atrial fibrillation	5 mg twice daily (or reduced 2.5 mg)
VTE treatment	10 mg twice daily x 7 days, then 5 mg twice daily [3]

Safety and monitoring

Specific reversal agent is **andexanet alfa** for life-threatening bleeding. Avoid with strong dual CYP3A4 and P-glycoprotein inhibitors or inducers [3].

CLINICAL PEARL Unlike warfarin, apixaban needs no routine coagulation monitoring, but adherence matters because its effect fades within ~24 hours of a missed dose.

3. Metformin

Biguanide — first-line for type 2 diabetes

KEY FACT Hold metformin before and for 48 hours after iodinated contrast when eGFR is below 30 mL/min/1.73m², due to lactic acidosis risk [4].

Mechanism of action

Metformin lowers hepatic gluconeogenesis and improves peripheral insulin sensitivity. It does not cause hypoglycaemia when used alone [4].

Dosing

Setting	Dose
Start	500 mg once or twice daily with meals
Maximum	2000-2550 mg/day in divided doses [4]

Safety and monitoring

Long-term use can cause vitamin **B12 deficiency**; check levels periodically. The most serious rare adverse effect is lactic acidosis, chiefly in renal impairment [4].

CLINICAL PEARL Taking metformin with food and titrating slowly reduces the common gastrointestinal side effects.

4. Atorvastatin

HMG-CoA reductase inhibitor (statin)

KEY FACT Atorvastatin 40–80 mg daily is a high-intensity statin, lowering LDL cholesterol by roughly ≥ 50% [5].

Mechanism of action

Statins inhibit HMG-CoA reductase, the rate-limiting enzyme of cholesterol synthesis, upregulating hepatic LDL receptors and clearing LDL from blood [5].

Dosing

Setting	Dose
High-intensity	40–80 mg once daily
Moderate-intensity	10–20 mg once daily [5]

Safety and monitoring

Monitor for **myopathy**; check creatine kinase if muscle pain occurs. Risk rises with strong CYP3A4 inhibitors such as clarithromycin. Obtain baseline liver enzymes [5].

CLINICAL PEARL Atorvastatin can be taken at any time of day, unlike shorter-acting statins that are dosed in the evening.

5. Lisinopril

Angiotensin-converting enzyme (ACE) inhibitor

KEY FACT ACE inhibitors are **contraindicated in pregnancy** (all trimesters) because of fetal renal toxicity [6].

Mechanism of action

Lisinopril blocks conversion of angiotensin I to angiotensin II, lowering blood pressure and reducing cardiac afterload. It also slows progression of diabetic kidney disease [6].

Dosing

Setting	Dose
Hypertension start	10 mg once daily
Heart failure	Start 2.5–5 mg daily, titrate [6]

Safety and monitoring

Monitor **serum potassium and creatinine** within 1–2 weeks of starting or increasing the dose. A dry cough and, rarely, angioedema are class effects [6].

CLINICAL PEARL A small rise in creatinine (up to ~30%) after starting an ACE inhibitor is expected and usually does not require stopping the drug.

6. Amoxicillin–clavulanate

Aminopenicillin + beta-lactamase inhibitor

KEY FACT The clavulanate component extends coverage to many **beta-lactamase–producing** organisms that plain amoxicillin would miss [7].

Mechanism of action

Amoxicillin inhibits bacterial cell-wall synthesis; clavulanic acid inactivates beta-lactamase enzymes that would otherwise destroy the amoxicillin [7].

Dosing

Setting	Dose
Adult (standard)	875 mg / 125 mg twice daily
Adult (high dose)	2000 mg / 125 mg twice daily [7]

Safety and monitoring

Do not use in patients with a documented **penicillin allergy**. Diarrhoea is the most common adverse effect, partly from the clavulanate [7].

CLINICAL PEARL The clavulanate dose is capped to limit gastrointestinal upset, which is why high-dose regimens raise only the amoxicillin amount.

7. Levothyroxine

Synthetic thyroid hormone (T4)

KEY FACT Full replacement dosing is approximately **1.6 mcg/kg/day**; older patients and those with cardiac disease start lower [8].

Mechanism of action

Levothyroxine is synthetic T4, converted peripherally to active T3. It restores normal thyroid hormone levels in hypothyroidism [8].

Dosing

Setting	Dose
Healthy adult	~1.6 mcg/kg once daily
Elderly / cardiac	Start 25–50 mcg daily [8]

Safety and monitoring

Take on an **empty stomach**, 30–60 minutes before food. Recheck **TSH in 6–8 weeks** after any dose change, since the hormone has a long half-life [8].

CLINICAL PEARL Calcium, iron, and antacids bind levothyroxine in the gut; separate them by at least 4 hours.

8. Drug Interactions and Monitoring

Cross-cutting safety reference

KEY FACT The combination of **warfarin + amiodarone** can sharply raise the INR and is a frequent cause of over-anticoagulation [2].

Mechanism of action

Many serious interactions act through the cytochrome P450 system or shared elimination pathways. The pairs below are among the most clinically important to anticipate [2][5].

Dosing

Setting	Dose
warfarin + amiodarone	INR rises; reduce warfarin, monitor closely [2]
atorvastatin + clarithromycin	Myopathy risk; hold or switch statin [5]
metformin + iodinated contrast	Lactic acidosis if renal impairment; hold metformin [4]

Safety and monitoring

When starting any interacting drug, plan the monitoring **before** the first dose: which lab, how soon, and what threshold triggers a change.

CLINICAL PEARL ACE inhibitors plus potassium-sparing diuretics or potassium supplements can cause dangerous hyperkalaemia [6].

References

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